

The Lens Opacities Classification System III Grading in Irradiated Uveal Melanomas to Characterize Proton Therapy-Induced Cataracts



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• **PURPOSE:** To evaluate the use of the Lens Opacities Classification System III grading (LOCS III) for the characterization of radiation-induced cataract, and to correlate the proton beam projection onto the lens with cataract location and grade as defined by the LOCS III.

• **DESIGN:** Prospective, interventional case series.

• **METHODS:** Fifty-two consecutive patients with cataract following proton therapy were included. All cataracts were graded using LOCS III. Relationships between proton beam and cataract subtypes, as well as between dose, proportion of lens irradiated, and extent of cataracts, were assessed.

• **RESULTS:** Tumor diameter, volume, stage, and equatorial tumor location were associated with extent of posterior subcapsular cataracts (PSC) that were diagnosed at a median (interquartile range) 36 months (22;83) after treatment. In multivariate analysis, the tumor volume ($P < .01$) and an equatorial tumor location ($P = .01$) were risk factors for extensive PSC. Lens irradiation was avoided in 10 patients. In the remaining 42 patients (81%), the extent of PSC significantly correlated with the dose to the lens receiving 10, 26, and 47 Gy ($P = .03$, $P = .03$, and $P = .04$, respectively), the dose to the lens periphery receiving 10 and 26 Gy ($P = .02$ and $P = .02$, respectively), and the dose to the ciliary body receiving 10 and 26 Gy ($P = .03$ and $P = .02$, respectively). Nuclear color significantly correlated with the dose to the ciliary body receiving 10 Gy ($P = .03$) and 26 Gy ($P = .02$). After adjustment of the results on tumor volume and tumor location, the volume of lens receiving

10 Gy ($P = .04$) and 26 Gy ($P = .03$) remained significantly associated with the extent of PSC.

• **CONCLUSIONS:** Proton dose correlated with the occurrence of PSC and nuclear color cataracts as defined by LOCS III grading. Better characterization of cataracts with the LOCS III after irradiation may help to further fill gaps in the current understanding of the mechanisms of radiation-induced cataracts. (Am J Ophthalmol 2019;201:63–71. © 2019 Elsevier Inc. All rights reserved.)

MANAGEMENT OF UVEAL MELANOMAS IS INCREASINGLY conservative, and proton beam therapy (PBT) and brachytherapy are the most frequently used techniques.^{1,2} However, radiation-induced toxicities such as optic neuropathy, retinopathy/maculopathy, dry eye syndrome, and cataracts can compromise vision.^{3–5} As the lens is one of the most radiosensitive tissues among the ocular structures, cataracts are a frequent complication after ocular radiation therapy. Their incidence varies from 45% to 100% after ocular radiation therapy.^{1,6,7} Radiation cataracts typically include posterior subcapsular lens clouding caused by damage to the subcapsular epithelium. However, there is neither absolute phenotypic conformity for the subtype of cataracts nor pathognomonic signs of radiation-induced cataracts. Senile, traumatic, or iatrogenic cataracts may also occur after radiation therapy. Full analysis of how exposure to therapeutic ionizing irradiation correlates and colocalizes with subsequent lens opacification is warranted to better establish causality. Such colocalization between radiation beam and opacities is only possible with specific forms of radiation modalities, in particular particle therapy that yields sharp dose distributions. Another important gap in current knowledge is that radiation cataracts are not graded like nononcologic cataracts.⁷ The international Common Toxicity Criteria (CTC) classification does provide grades of severity in terms of visual acuity and cataracts but is not appropriate for uveal melanomas, as it does not distinguish between vision loss owing to maculopathy or cataracts and does not describe lens opacities accurately. Depending on their location in the lens, opalescence, color, and proportion of lens involved, lens opacifications can variably affect the vision. The international Lens Opacity Classification System (LOCS) defines a score based on the anatomic location and extent of cataracts on slit-lamp

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